Multimodal, Interactive Systems, UXD-M
Exercise, Summer term 2027

Exercise Sheet 04
Prof. Dr. Simon Nestler

One prototype, two modalities

What is this sheet about?

This sheet carries the big milestone of the semester: one working multimodal prototype. The gesture work from block 2 and the dialog work from block 3 have to run in one system for one task, with defined handoffs between the channels. Working means that someone outside your team can operate it without help. Rough and unfinished is allowed.

Docking point: This exercise cashes in sessions T06 (*Anchor I: The car case*) and T08 (*Chat can ask anything*): the anchor case taught you that a modality architecture is a set of justified channel assignments with a price per decision, and T08 gave you the criteria for when conversation belongs in it at all — today your own project has to survive both.

By the end of this block your team has the milestone from the semester plan, and it is the big one: a **working multimodal prototype**. Concretely: your gesture work from exercise block 2 and your dialog work from exercise block 3 (or the two modalities your project idea names) run in *one* system for *one* task, with defined handoffs between the channels. "Working" means: someone outside your team can operate it without a team member's hands hovering over the keyboard. It may be rough and unfinished.

Multimodal, Interactive Systems, UXD-M
Exercise, Summer term 2027

Exercise Sheet 04
Prof. Dr. Simon Nestler

Organisation: Please work on this exercise sheet in your project team (fixed since exercise block 1). Within this sheet the team members are called A, B, C (and D, if you are working in a team of four). For the build phase, one team member acts as the *integration owner* who holds the whole system; the others each own a channel. In order to avoid confusions please document at the beginning who will take which (generic) role.

AI role: tool (declared). Generate glue code, screens, and assets as you like; you carry responsibility for every behavior in the demo. Your upload **must disclose what was made with which tool**.

Bonus points: In order for the bonus rule to apply (participation in all six blocks with an upload within 48 hours each earns 5 % compensation on the exam), your team must hand in one complete submission package (see the end of this sheet) within 48 hours after this exercise. This is block four of six.

Tools: *ProtoPie* as the likely integration hub (touch, voice triggers, and external inputs in one project); *MediaPipe Gesture Recognizer* or your *Teachable Machine* model from exercise block 2 for the gesture channel; *Voiceflow* or ProtoPie voice for the conversational channel, if your project has one; *Figma Make* for screens you still owe yourself. Teams building for headset or 3D contexts may use *Unity* with Android XR support or *ShapesXR* instead. Declare it in the submission; the milestone is the same. Wiring two half-working things together is harder than building either half. Plan for the integration to fight back.

First assignment (20 minutes): Write an integration plan: one page, as a team, written *before* building. Which user intention goes through which channel, where does a handoff happen (gesture starts, voice confirms, or the reverse), and where could the two channels contradict each other? Use the T02 triad explicitly: per channel pairing, is it redundancy, complementarity, or a conflict waiting to fire? Name the price of your assignment, T06-style. Then walk your task step by step and mark, per step, whether the intention is carried redundantly by two channels or by one channel alone. For every single-channel step, name who loses access when that channel fails — situationally (noise, gloves, sunlight), temporarily (a broken arm, a lost voice) or permanently (a person who is blind, deaf, or cannot speak); these impairments come in degrees, rarely as all or nothing. This is the *Design for All* check of EN 17161.

Second assignment (75 minutes): Build the integrated prototype as a team, with split roles: the integration owner holds the whole, the others work on channels. Work the plan, and log every deviation from it with one sentence of reason. That log is a submission artifact, because the distance between plan and build is where you will learn the most.

Third assignment (25 minutes): Transition test, in pairs across teams. Swap a tester with your neighbor team. The foreign tester operates your prototype through one full task, deliberately **switching modality mid-task at least twice**. You observe and protocol: where does a handoff drop state, feedback, or the user? Protocol each handoff as one row with four columns: *perceivable?* (does the tester notice the new channel is active), *identifiable?* (do they know what to do in it), *feedback?* (does the system confirm the handoff), *completable?* (do they get through without help). Add a comment and a screenshot or timestamp. Three findings minimum. Mode confusion at transitions is precisely the LZ2 exam material, found live in your own build.

Fourth assignment (20 minutes): Demo preparation, as a team. Fix the one transition finding you can fix in twenty minutes; record a continuous demo of the full task (**one take, no cuts**: a cut hides exactly what critique needs to see).

Multimodal, Interactive Systems, UXD-M
Exercise, Summer term 2027

Exercise Sheet 04
Prof. Dr. Simon Nestler

Fifth assignment (10 minutes): Upload preparation, as a team. Package the submission (see below).

Critique round 1 (30 minutes, plenum): This round uses the buffer time, and every team takes part. Each team: two minutes of demo, then three minutes of structured critique from the room along the questions from the first assignment. Channel assignment justified? Handoffs holding? Price named? The presenting team answers each objection with exactly one of: *accepted* or *disputed with evidence*. No free-form defense; the format is rehearsal for exercise block 6 and for T12's cross-critique.

Submission (within 48 h, one upload per team):

- `prototype/` — the integrated prototype: ProtoPie file, Voiceflow link, code, or Unity/ShapesXR project reference, with a `run.md` stating exactly how to start it.
- `integration-plan.md` — the plan from task 1 plus the deviation log from task 2 (max 2 pages).
- `transition-findings.md` — the three-plus findings from the foreign tester, each with its handoff location (max 1 page).
- Demo recording, one take (max 3 minutes, .mp4, max 200 MB).
- `critique-notes.md` — the objections your team received in critique round 1, each marked *accepted* or *disputed with evidence* (max 1 page).
- **AI disclosure line** (in `integration-plan.md`, mandatory): *"Made with AI: [tool] was used for [part]; checked and understood by [name]."*

Put everything into one upload per team and hand it in via Moodle within 48 hours after this exercise.